NEWTON'S LAWS OF MOTION - Revised Core 2

20 source-demand transfer challenges with paired prompt/reveal hints

REVISED CORE 2

Attempt first. Each hint is a question before it is a reveal. H1 asks for the decisive physics, H2 asks for the representation, and H3 asks for the first mathematical or logical constraint. The reveal must answer only that rung.

H1 - What is the key physics?

Event, state, force/body ownership, system boundary, direction, or invariant.

H2 - How would you represent it?

FBD, interaction pair, momentum state, graph, system boundary, or signed axis.

H3 - What would you write first?

One starting equation/constraint only; no final-result leakage.

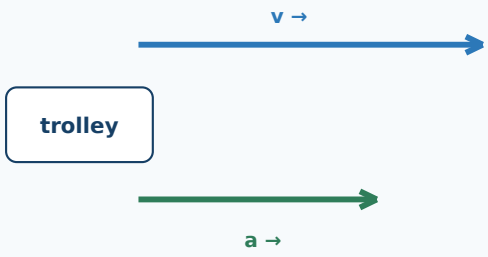
Q01 - Changing velocity means net force

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CHALLENGE

QUESTION

A trolley moves to the right and its speed increases steadily. Which statement about the horizontal net force is necessarily true?



HINT 1 - WHAT IS THE KEY PHYSICS?

The trolley is moving right and speeding up. What does that tell you about the direction of its acceleration?

REVEAL

Its acceleration points to the right, the same direction as the velocity while the speed increases.

HINT 2 - HOW WOULD YOU REPRESENT IT?

For one chosen trolley, what representation connects that acceleration to the forces acting on it?

REVEAL

Draw the trolley FBD and form the horizontal vector sum ΣF_x.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What law would you write first after determining the acceleration direction?

REVEAL

Write $\Sigma F_x = ma_x$. Use the acceleration direction from H1 only after you have written the law.

YOUR WORK

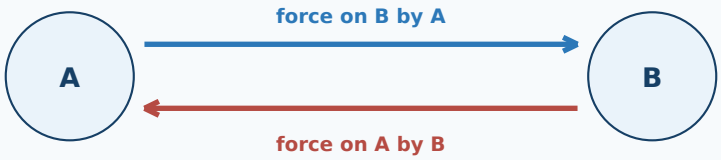
Q02 - Identify the Newton-III pair

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CHALLENGE

QUESTION

A student pushes a wall horizontally with 120 N. What force does the wall exert on the student?



HINT 1 - WHAT IS THE KEY PHYSICS?

The student and wall are one interaction pair. What does Newton III say about the two interaction forces?

REVEAL

They are equal in magnitude, opposite in direction, and act on different bodies.

HINT 2 - HOW WOULD YOU REPRESENT IT?

How would you label the pair so body ownership cannot be confused?

REVEAL

Use “force on wall by student” and “force on student by wall”.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

If the student-on-wall force is +120 N, what relation would you write first for the wall-on-student force?

REVEAL

F_student by wall = -F_wall by student.

YOUR WORK

Q03 - Catching time and average force

Attempt page - answer protected

CHALLENGE

QUESTION

A 0.40 kg ball moving at 15 m/s is brought to rest. Compare the average force magnitude if it is stopped in 0.10 s versus 0.30 s.



HINT 1 - WHAT IS THE KEY PHYSICS?

The initial and final momenta are the same in both catches. What quantity changes between the two cases?

REVEAL

Only the stopping time changes; the momentum change is the same.

HINT 2 - HOW WOULD YOU REPRESENT IT?

Which state representation makes the sign and size of the momentum change explicit?

REVEAL

Write $p_i = mv_i$, $p_f = 0$, then $\Delta p = p_f - p_i$ for each case.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What force relation would you write first once Δp is known?

REVEAL

$F_{avg} = \Delta p / \Delta t$.

YOUR WORK

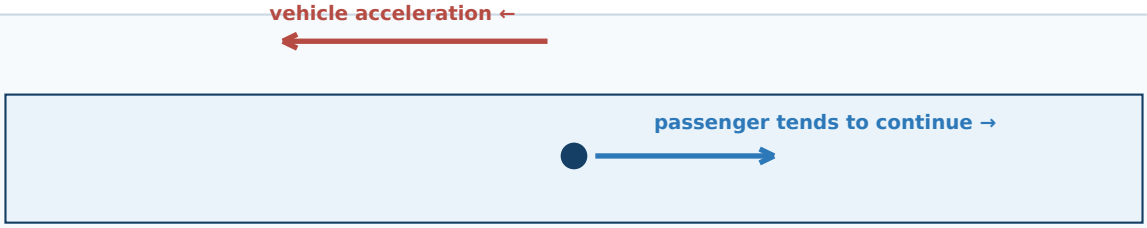
Q04 - Inertia during sudden braking

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CHALLENGE

QUESTION

A standing passenger in a bus lurches forward when the bus brakes suddenly. Explain the effect without inventing a forward force of inertia.



HINT 1 - WHAT IS THE KEY PHYSICS?

Immediately before braking, what horizontal velocity does the passenger's upper body already have?

REVEAL

It has the same forward velocity as the bus.

HINT 2 - HOW WOULD YOU REPRESENT IT?

When the bus floor slows the passenger's feet, what representation separates the foot-contact effect from the upper body's tendency to continue?

REVEAL

Compare the passenger parts/body state with the braking bus: contact changes the supported part first while the upper body retains forward velocity momentarily.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

Which Newton-law idea would you state first?

REVEAL

Newton I: without enough backward external impulse, the body tends to retain its current velocity.

YOUR WORK

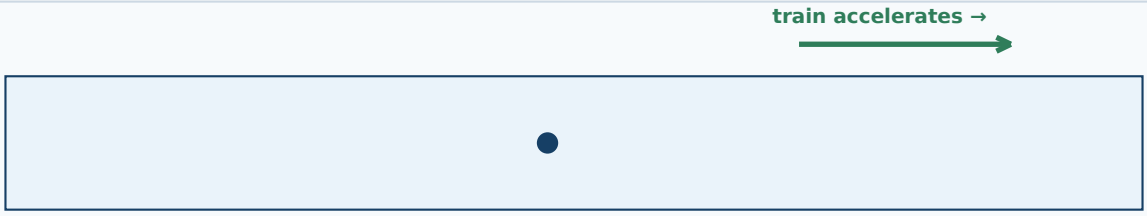
Q05 - Coin tossed in an accelerating train

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CHALLENGE

QUESTION

Inside a train accelerating forward, a passenger tosses a coin vertically. Relative to the passenger, where does the coin tend to land?



Airborne coin keeps its prior horizontal velocity while train gains speed.

HINT 1 - WHAT IS THE KEY PHYSICS?

While the coin is airborne, does it automatically share the train's continuing forward acceleration?

REVEAL

No. Once released, it retains its horizontal velocity at release unless a horizontal force acts.

HINT 2 - HOW WOULD YOU REPRESENT IT?

How would you compare train and coin horizontal motion during the same flight time?

REVEAL

Draw two horizontal state tracks: the train gains forward speed; the coin keeps approximately constant horizontal speed.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What relative-motion statement would you make first at landing time?

REVEAL

Compare x_train(t) and x_coin(t) at the same landing time; the sign of x_coin-x_train gives the relative landing position.

YOUR WORK

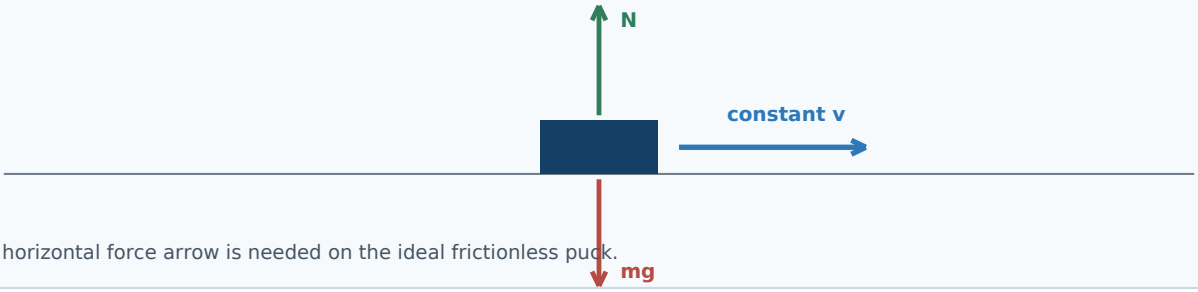
Q06 - Constant velocity on a frictionless table

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CHALLENGE

QUESTION

A puck slides at constant velocity on a perfectly frictionless horizontal table. What horizontal force is needed to keep it moving?



No horizontal force arrow is needed on the ideal frictionless puck.

HINT 1 - WHAT IS THE KEY PHYSICS?

The puck's velocity is constant. What is its acceleration?

REVEAL

Zero.

HINT 2 - HOW WOULD YOU REPRESENT IT?

What does a one-body FBD say about the horizontal net force when a=0?

REVEAL

The horizontal vector sum must be zero.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

Which law connects that conclusion to motion?

REVEAL

Write $\Sigma F_x = ma_x$. Then use the constant-velocity condition to supply a_x.

YOUR WORK

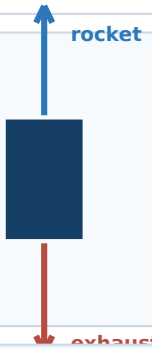
Q07 - Rocket propulsion and momentum

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CHALLENGE

QUESTION

A rocket ejects exhaust backward in deep space. Explain why the rocket accelerates forward.



HINT 1 - WHAT IS THE KEY PHYSICS?

Which two parts exchange momentum during the short ejection event?

REVEAL

Rocket and exhaust.

HINT 2 - HOW WOULD YOU REPRESENT IT?

What system boundary lets the internal exchange be seen without an external horizontal/axial force?

REVEAL

Choose rocket + expelled exhaust over the short ejection interval.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What momentum relation would you write first for an initially at-rest simplified system?

REVEAL

0 = p_rocket + p_exhaust, with opposite signs.

YOUR WORK

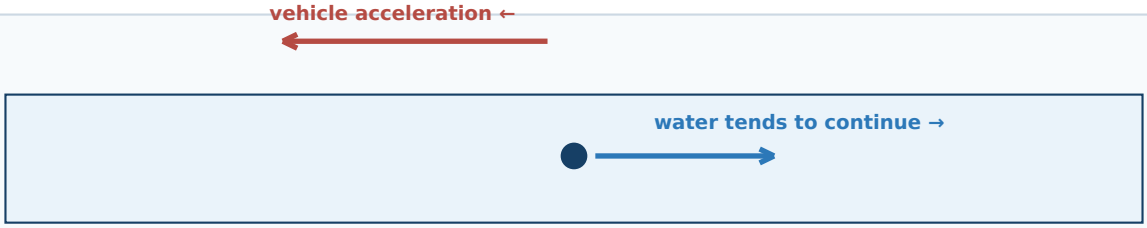
Q08 - Water tanker braking and inertia

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CHALLENGE

QUESTION

A partly filled tanker brakes suddenly. Why does the water surge toward the front?



HINT 1 - WHAT IS THE KEY PHYSICS?

Before braking, what velocity does the water already have relative to the road?

REVEAL

Approximately the tanker’s forward velocity.

HINT 2 - HOW WOULD YOU REPRESENT IT?

When the tanker body decelerates, what physical comparison should you make between the container and the water?

REVEAL

The container is forced to slow by the brakes/road; the water tends to retain its forward velocity until wall/fluid forces change it.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

Which Newton-law idea would you state first?

REVEAL

Newton I: matter resists a change of velocity.

YOUR WORK

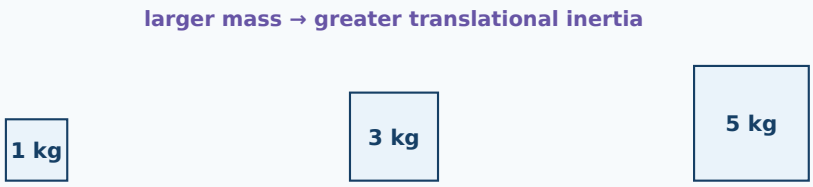
Q09 - Mass as a measure of inertia

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CHALLENGE

QUESTION

Three blocks have the same shape but masses 1 kg, 3 kg and 5 kg. Rank their inertia.



HINT 1 - WHAT IS THE KEY PHYSICS?

What property measures resistance to a change in velocity in translational motion?

REVEAL

Mass.

HINT 2 - HOW WOULD YOU REPRESENT IT?

How would you represent the comparison without using a force calculation?

REVEAL

Use a simple mass ranking: larger mass → greater inertia.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What inequality would you write first?

REVEAL

Write the mass order first: 5 kg > 3 kg > 1 kg. Then state separately how inertia tracks mass.

YOUR WORK

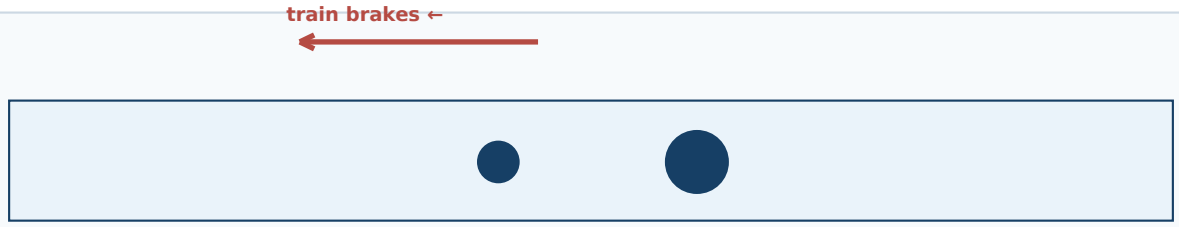
Q10 - Ambiguous rolling claim on a smooth train floor

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CHALLENGE

QUESTION

A source question places balls on a “smooth” train floor during braking and asks about rolling. Under the literal ideal-frictionless interpretation, what can and cannot be concluded?



Ideal smooth floor: vertical N and mg only; no horizontal frictional torque.

HINT 1 - WHAT IS THE KEY PHYSICS?

If the floor is truly frictionless, what horizontal contact force and rolling torque can it exert on a ball?

REVEAL

No horizontal friction force and therefore no frictional rolling torque.

HINT 2 - HOW WOULD YOU REPRESENT IT?

What representation separates translational inertia from rotational rolling?

REVEAL

Draw the ball with vertical N and mg only; there is no horizontal friction arrow in the ideal smooth-floor model.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What conclusion would you write first before discussing the source’s intended inertia idea?

REVEAL

Start with $\Sigma F_x = 0$ and $\tau_{contact} = 0$ for the literal ideal-frictionless contact. Only then decide what motion is supported.

YOUR WORK

Q11 - Light and heavy rifle recoil

Attempt page - answer protected

CHALLENGE

QUESTION

Two rifles fire bullets with the same forward momentum. One rifle has half the mass of the other. Compare their recoil momenta and recoil speeds.



HINT 1 - WHAT IS THE KEY PHYSICS?

For rifle+bullet systems initially at rest, what must the rifle momentum do when the bullet gets forward momentum?

REVEAL

It must have equal-magnitude opposite momentum, if external impulse is negligible during firing.

HINT 2 - HOW WOULD YOU REPRESENT IT?

If both rifles have the same recoil momentum magnitude, which quantity controls their recoil speed?

REVEAL

v = p/M, so mass controls the speed for fixed momentum.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What ratio would you write first for the recoil speeds?

REVEAL

v_light/v_heavy = M_heavy/M_light.

YOUR WORK

Q12 - Horse and cart at constant speed

Attempt page - answer protected

CHALLENGE

QUESTION

A horse pulls a cart at constant speed on a level road. The horse pulls forward on the cart with 500 N. What must the total backward resistance on the cart be?



HINT 1 - WHAT IS THE KEY PHYSICS?

The cart moves at constant velocity. What does that say about its acceleration and net force?

REVEAL

Both are zero.

HINT 2 - HOW WOULD YOU REPRESENT IT?

Which body should the FBD contain if the question asks about resistance on the cart?

REVEAL

The cart only: forward horse-on-cart force and backward resistive forces.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What horizontal balance equation would you write first?

REVEAL

$\Sigma F_x = 500 - R = 0$.

YOUR WORK

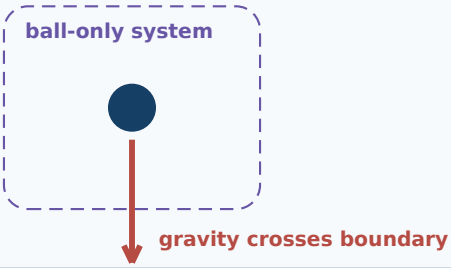
Q13 - Why a falling ball alone does not conserve momentum

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CHALLENGE

QUESTION

A ball falls freely and its downward momentum increases. Can you say the ball’s momentum is conserved because the initial and final momentum happen to have some matching magnitude in two observations?



HINT 1 - WHAT IS THE KEY PHYSICS?

What external force acts on the ball-only system during the fall?

REVEAL

Gravity.

HINT 2 - HOW WOULD YOU REPRESENT IT?

How would a system boundary around only the ball affect the conservation test?

REVEAL

Gravity crosses the boundary as an external interaction and gives external impulse.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What conservation check must be written before comparing momenta?

REVEAL

Write the conservation test first: is the external impulse on the chosen ball-only system negligible over the interval?

YOUR WORK

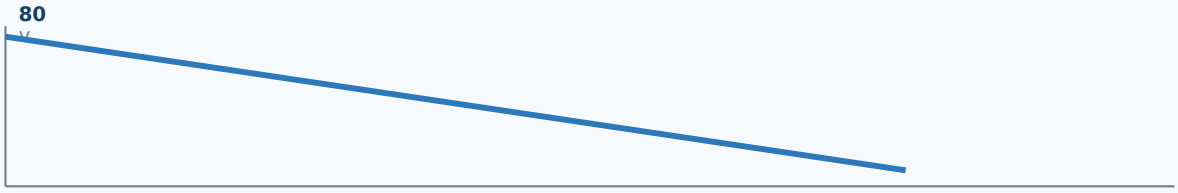
Q14 - Friction from a velocity-time graph

Attempt page - answer protected

CHALLENGE

QUESTION

A 50 g slider slows uniformly from 80 m/s to rest in 8 s. If friction is the only horizontal force, find the friction force.



slope = acceleration → friction force if friction is the only horizontal force

HINT 1 - WHAT IS THE KEY PHYSICS?

The graph is velocity versus time. What physical quantity is the slope?

REVEAL

Acceleration.

HINT 2 - HOW WOULD YOU REPRESENT IT?

How would you represent the mass before using Newton II?

REVEAL

Convert 50 g to 0.050 kg and use the signed graph slope for a.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What slope would you write first?

REVEAL

a=(0-80)/(8-0).

YOUR WORK

Q15 - Force-mass scaling

Attempt page - answer protected

CHALLENGE

QUESTION

An object accelerates at a under net force F and mass m. The force becomes F/2 while the mass becomes 2m. What is the new acceleration?

state	Fnet	mass	acceleration
old	F	m	a
new	F/2	2m	?

HINT 1 - WHAT IS THE KEY PHYSICS?

Both numerator and denominator of a=F/m change. Which way does each change push the acceleration?

REVEAL

Halving force halves a; doubling mass halves it again.

HINT 2 - HOW WOULD YOU REPRESENT IT?

How would you represent the old and new states without assigning numbers?

REVEAL

$a_1=F/m$ and $a_2=(F/2)/(2m)$.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What ratio would you write first?

REVEAL

$a_2/a_1=(1/2)(1/2)$.

YOUR WORK

Q16 - Skaters push apart

Attempt page - answer protected

CHALLENGE

QUESTION

Two skaters of masses 50 kg and 75 kg push apart from rest. The 50 kg skater moves left at 3 m/s. Find the velocity of the 75 kg skater.



Check external impulse before conserving total momentum.

HINT 1 - WHAT IS THE KEY PHYSICS?

For the two-skater system initially at rest, what is the total momentum before they push?

REVEAL

Zero.

HINT 2 - HOW WOULD YOU REPRESENT IT?

How would you encode opposite directions so momentum conservation can be used without losing direction?

REVEAL

Choose right positive: v_1=-3 m/s for the left-moving 50 kg skater; v_2 is unknown.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What momentum equation would you write first?

REVEAL

$0 = 50(-3) + 75v_2$.

YOUR WORK

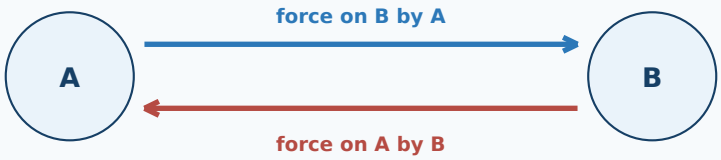
Q17 - Sprinkler reaction

Attempt page - answer protected

CHALLENGE

QUESTION

A sprinkler ejects water tangentially backward from its nozzles. Why does the sprinkler rotate in the opposite sense?



HINT 1 - WHAT IS THE KEY PHYSICS?

What body exerts the force that accelerates the outgoing water?

REVEAL

The sprinkler/nozzle exerts a force on the water.

HINT 2 - HOW WOULD YOU REPRESENT IT?

What is the Newton-III partner of that force and on which body does it act?

REVEAL

Water exerts an equal-and-opposite force on the sprinkler.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What directional statement would you write first for the pair?

REVEAL

F_on sprinkler by water = -F_on water by sprinkler.

YOUR WORK

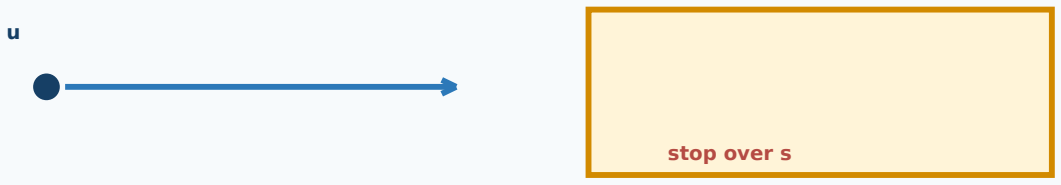
Q18 - Bullet stopping force and time

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CHALLENGE

QUESTION

A 0.020 kg bullet enters a block at 100 m/s and stops uniformly in 0.10 m. Find the average stopping force and stopping time.



HINT 1 - WHAT IS THE KEY PHYSICS?

Distance is given but acceleration is not. Which model must be used before Newton II?

REVEAL

Constant-acceleration kinematics.

HINT 2 - HOW WOULD YOU REPRESENT IT?

What motion state would you write so the acceleration can be found without time?

REVEAL

u=100 m/s, v=0, s=0.10 m; use $v^2=u^2+2as$.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What equation would you write first?

REVEAL

$0 = 100^2 + 2a(0.10)$.

YOUR WORK

Q19 - Combined masses under the same force

Attempt page - answer protected

CHALLENGE

QUESTION

A 5 N force gives mass m_1 acceleration 8 m/s² and mass m_2 acceleration 24 m/s². If the masses are tied together and the same 5 N net force acts, find the acceleration.

case	F	a	m=F/a
1	5 N	8 m/s²	5/8 kg
2	5 N	24 m/s²	5/24 kg

HINT 1 - WHAT IS THE KEY PHYSICS?

The same force produces different accelerations. What does F=ma let you infer about each mass?

REVEAL

m=F/a, so the smaller acceleration belongs to the larger mass.

HINT 2 - HOW WOULD YOU REPRESENT IT?

How would you represent the tied system before finding its new acceleration?

REVEAL

Find m_1 and m_2 separately, then add them because both are accelerated together by the same net force.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What mass equations would you write first?

REVEAL

m_1=5/8 and m_2=5/24 kg.

YOUR WORK

Q20 - Momentum graphs and force as rate of change

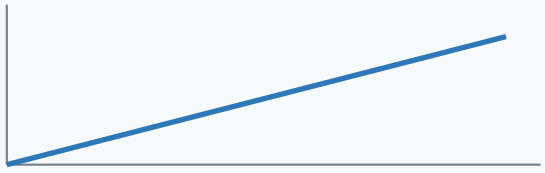
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CHALLENGE

QUESTION

For p=mv, what do the slopes of a p-v graph at fixed mass and a p-m graph at fixed velocity represent? How is force related to momentum change?

p vs v



slope = mass

p vs m



slope = velocity

HINT 1 - WHAT IS THE KEY PHYSICS?

In p=mv, if m is held fixed and v varies, which quantity multiplies the horizontal-axis variable?

REVEAL

Mass; it is the slope of p versus v.

HINT 2 - HOW WOULD YOU REPRESENT IT?

If velocity is held fixed and mass varies, what does the corresponding graph slope become?

REVEAL

Velocity; p=vm is linear in m.

HINT 3 - WHAT WOULD YOU WRITE FIRST?

What rate equation would you write first for force in momentum language?

REVEAL

$F_{avg} = \Delta p/\Delta t$ (and instantaneously $F=dp/dt$).

YOUR WORK

Quick Answer Key

Results only - use after attempting

CHECK

Q01 Net horizontal force points right.

Q02 120 N opposite to the push.

Q03 60 N versus 20 N; the 0.30 s catch has one-third the force.

Q04 Passenger lurches forward because the body tends to retain its pre-braking forward velocity.

Q05 Behind the passenger.

Q06 0 N.

Q07 Forward rocket momentum balances backward exhaust momentum in the simplified system.

Q08 Water surges toward the front.

Q09 5 kg > 3 kg > 1 kg.

Q10 Rolling is not justified by the literal smooth-floor model; the inertia comparison is the usable idea.

Q11 Equal recoil momentum; lighter rifle has twice the recoil speed.

Q12 500 N backward.

Q13 No; gravity gives external impulse to the ball-only system.

Q14 0.50 N opposite the motion.

Q15 a/4.

Q16 2 m/s to the right.

Q17 Opposite to the water jets.

Q18 1000 N opposite motion; 0.0020 s.

Q19 6 m/s^2.

Q20 slope(p-v)=m; slope(p-m)=v; $F_{avg}=\Delta p/\Delta t$.

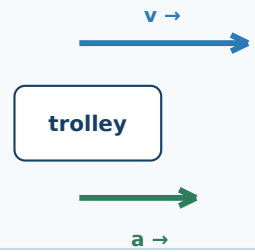
Q01 - Solution: Changing velocity means net force

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A trolley moves to the right and its speed increases steadily. Which statement about the horizontal net force is necessarily true?



WHAT DO WE KNOW?

body: trolley; motion: rightward and speeding up.

REASONING LADDER

1 UNDERSTAND

The velocity magnitude is increasing to the right, so acceleration is rightward.

2 REPRESENT

Use one trolley FBD and the horizontal axis.

3 CONNECT

Newton II requires $\Sigma F_x = ma_x$.

4 CALCULATE

Since $a_x > 0$, $\Sigma F_x > 0$: the net horizontal force is rightward.

5 INTERPRET

A rightward net force is required; individual forces need not all point right.

ANSWER

Net horizontal force points right.

QUICK CHECK

If the net force were zero, the velocity could not be changing.

CHECK IT ANOTHER WAY

A leftward net force would make a right-moving trolley slow down, not speed up.

WATCH OUT

Do not infer force direction from velocity alone; use the change of velocity.

REVIEW THIS IDEA

Core 1: NLM 01-04

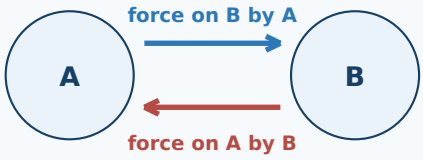
Q02 - Solution: Identify the Newton-III pair

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A student pushes a wall horizontally with 120 N. What force does the wall exert on the student?



WHAT DO WE KNOW?

interaction: student ↔ wall; applied force magnitude 120 N.

REASONING LADDER

1 UNDERSTAND

This is one contact interaction between two bodies.

2 REPRESENT

Represent the pair on two different bodies.

3 CONNECT

Newton III: F(on student by wall) = -F(on wall by student).

4 CALCULATE

Magnitude is therefore 120 N, opposite the push direction.

5 INTERPRET

The wall exerts 120 N opposite to the student’s push.

ANSWER

120 N opposite to the push.

QUICK CHECK

The two forces are equal even if the wall barely accelerates.

CHECK IT ANOTHER WAY

The student acceleration depends on the net force on the student, not on “cancelling” this pair.

WATCH OUT

Do not put both third-law forces on the same FBD.

REVIEW THIS IDEA

Core 1: NLM 08

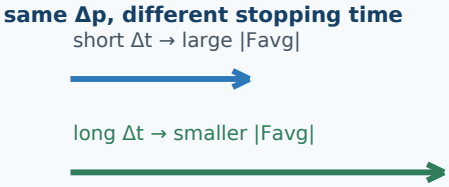
Q03 - Solution: Catching time and average force

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A 0.40 kg ball moving at 15 m/s is brought to rest. Compare the average force magnitude if it is stopped in 0.10 s versus 0.30 s.



WHAT DO WE KNOW?

m=0.40 kg; v_i=15 m/s; v_f=0; Δt=0.10 s or 0.30 s.

REASONING LADDER

1 UNDERSTAND

Same Δp in both catches; only Δt differs.

2 REPRESENT

Take motion as positive: p_i=0.40(15)=6 kg·m/s and p_f=0.

3 CONNECT

Use $F_{avg}=\Delta p/\Delta t$ with Δp=-6 kg·m/s.

4 CALCULATE

|F|=6/0.10=60 N; second case |F|=6/0.30=20 N.

5 INTERPRET

Tripling stopping time reduces average force magnitude to one-third.

ANSWER

60 N versus 20 N; the 0.30 s catch has one-third the force.

QUICK CHECK

60×0.10 = 20×0.30 = 6 N·s, the same impulse magnitude.

CHECK IT ANOTHER WAY

A longer stopping time with the same impulse must lower average force.

WATCH OUT

Do not compare force using speed alone; compare Δp and Δt.

REVIEW THIS IDEA

Core 1: NLM 06-07

Q04 - Solution: Inertia during sudden braking

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A standing passenger in a bus lurches forward when the bus brakes suddenly. Explain the effect without inventing a forward force of inertia.



WHAT DO WE KNOW?

bus initially moving; bus acceleration becomes backward during braking.

REASONING LADDER

1 UNDERSTAND

The passenger initially shares the bus forward velocity.

2 REPRESENT

Represent the bus as slowing beneath the passenger; avoid a fictitious forward force in an inertial-frame FBD.

3 CONNECT

Newton I explains persistence of velocity until external forces change it.

4 CALCULATE

Feet are decelerated by contact; the upper body continues forward briefly, producing the lurch.

5 INTERPRET

The forward lurch is an inertia effect, not evidence of a new forward force.

ANSWER

Passenger lurches forward because the body tends to retain its pre-braking forward velocity.

QUICK CHECK

If the bus braked gently over a longer time, the lurch would be smaller.

CHECK IT ANOTHER WAY

The actual contact force during braking is backward on the passenger, not forward.

WATCH OUT

Do not add a “force of inertia” to an ordinary inertial-frame FBD.

REVIEW THIS IDEA

Core 1: NLM 01-02

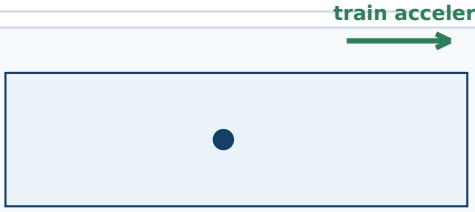
Q05 - Solution: Coin tossed in an accelerating train

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

Inside a train accelerating forward, a passenger tosses a coin vertically. Relative to the passenger, where does the coin tend to land?



Airborne coin keeps its prior horizontal velocity while train gains s

WHAT DO WE KNOW?

train accelerates forward; coin released with train's instantaneous horizontal velocity.

REASONING LADDER

1 UNDERSTAND

At release, coin and train share horizontal velocity; after release only the train continues accelerating forward.

2 REPRESENT

Compare their horizontal positions over the same airborne time.

3 CONNECT

Train displacement grows faster because of its additional $+\frac{1}{2}at^2$ term.

4 CALCULATE

Thus the passenger/train moves ahead of the coin horizontally.

5 INTERPRET

The coin lands behind the passenger (relative to the accelerating train).

ANSWER

Behind the passenger.

QUICK CHECK

If train acceleration were zero, the coin would return to the thrower horizontally.

CHECK IT ANOTHER WAY

The coin does not "lose" forward velocity; the train gains additional forward velocity.

WATCH OUT

Do not treat the coin as if it remains rigidly attached to the accelerating train.

REVIEW THIS IDEA

Core 1: NLM 01-02

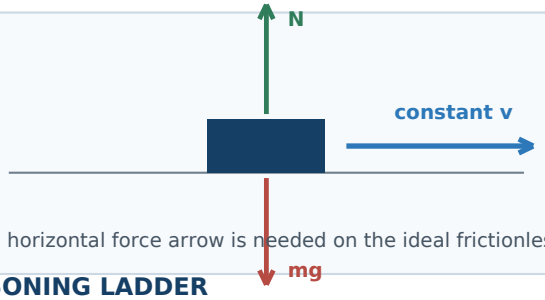
Q06 - Solution: Constant velocity on a frictionless table

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A puck slides at constant velocity on a perfectly frictionless horizontal table. What horizontal force is needed to keep it moving?



No horizontal force arrow is needed on the ideal frictionless puck.

WHAT DO WE KNOW?

constant horizontal velocity; ideal frictionless surface.

REASONING LADDER

1 UNDERSTAND

Constant velocity means zero acceleration.

2 REPRESENT

The vertical forces can balance; horizontally no force is required.

3 CONNECT

$\Sigma F_x = ma_x = 0$.

4 CALCULATE

Therefore required horizontal net force is 0 N.

5 INTERPRET

Motion persists without a sustaining horizontal force.

ANSWER

0 N.

QUICK CHECK

A nonzero horizontal force would change the velocity, contradicting the condition.

CHECK IT ANOTHER WAY

The table still exerts a vertical normal force; “zero horizontal force” is not “no forces at all”.

WATCH OUT

Do not confuse moving with accelerating.

REVIEW THIS IDEA

Core 1: NLM 01-03

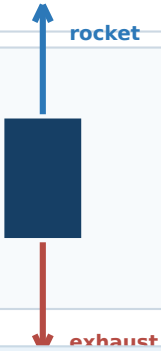
Q07 - Solution: Rocket propulsion and momentum

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A rocket ejects exhaust backward in deep space. Explain why the rocket accelerates forward.



WHAT DO WE KNOW?

system: rocket + expelled exhaust; external impulse neglected over a short interval.

REASONING LADDER

1 UNDERSTAND

Exhaust is accelerated backward; the rocket receives equal-and-opposite interaction forces and momentum change.

2 REPRESENT

Use a system boundary around rocket plus exhaust and choose forward positive.

3 CONNECT

Conserve total momentum for the simplified short interval.

4 CALCULATE

Backward exhaust momentum is negative; rocket momentum becomes positive to keep total momentum unchanged.

5 INTERPRET

The rocket accelerates forward while exhaust carries backward momentum.

ANSWER

Forward rocket momentum balances backward exhaust momentum in the simplified system.

QUICK CHECK

Newton III gives equal/opposite interaction forces; momentum conservation gives the corresponding opposite momentum changes.

CHECK IT ANOTHER WAY

The rocket does not need air to “push against”.

WATCH OUT

Do not explain rocket propulsion as exhaust pushing on external air.

REVIEW THIS IDEA

Core 1: NLM 08-10

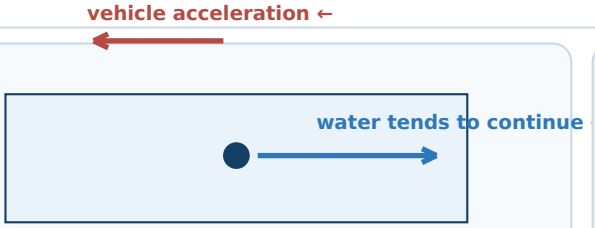
Q08 - Solution: Water tanker braking and inertia

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A partly filled tanker brakes suddenly. Why does the water surge toward the front?



WHAT DO WE KNOW?

tanker initially moving forward; tanker acceleration becomes backward.

REASONING LADDER

1 UNDERSTAND

Water has forward momentum before braking.

2 REPRESENT

Compare the braking tank wall with water that initially continues forward relative to the road.

3 CONNECT

Newton I explains the persistence of the water's forward motion.

4 CALCULATE

Relative to the decelerating tank, the water shifts/surges toward the front.

5 INTERPRET

The surge is an inertia effect caused by different rates of velocity change.

ANSWER

Water surges toward the front.

QUICK CHECK

Gentler braking over more time reduces the relative surge.

CHECK IT ANOTHER WAY

No forward "inertia force" is needed in a road-frame explanation.

WATCH OUT

Do not confuse relative motion in the tanker with an extra fundamental force.

REVIEW THIS IDEA

Core 1: NLM 01

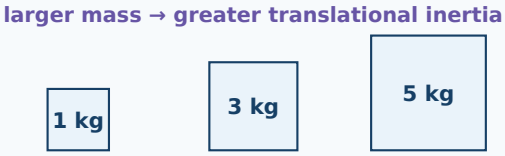
Q09 - Solution: Mass as a measure of inertia

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

Three blocks have the same shape but masses 1 kg, 3 kg and 5 kg. Rank their inertia.



WHAT DO WE KNOW?

masses: 1 kg, 3 kg, 5 kg.

REASONING LADDER

1 UNDERSTAND

Inertia in translational motion scales with mass.

2 REPRESENT

No FBD is needed for a pure ranking; compare masses directly.

3 CONNECT

Larger mass means greater resistance to the same acceleration change.

4 CALCULATE

Therefore inertia: 5 kg > 3 kg > 1 kg.

5 INTERPRET

The 5 kg block has the greatest inertia.

ANSWER

5 kg > 3 kg > 1 kg.

QUICK CHECK

Under the same net force, their accelerations would rank in the reverse order.

CHECK IT ANOTHER WAY

Inertia is not density or size by itself.

WATCH OUT

Do not rank inertia from volume when mass is given.

REVIEW THIS IDEA

Core 1: NLM 01, NLM 04

Q10 - Solution: Ambiguous rolling claim on a smooth train floor

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A source question places balls on a “smooth” train floor during braking and asks about rolling. Under the literal ideal-frictionless interpretation, what can and cannot be concluded?



Ideal smooth floor: vertical N and mg only; no horizontal frictional

WHAT DO WE KNOW?

literal model: ideal smooth/frictionless floor during train braking.

REASONING LADDER

1 UNDERSTAND

With no horizontal friction, the ball has no horizontal contact force and no frictional torque.

2 REPRESENT

One-ball FBD: N upward, mg downward, no horizontal friction.

3 CONNECT

Thus horizontal acceleration is zero in the ground frame; rolling cannot be inferred from the stated contact model.

4 CALCULATE

The train slows beneath the ball, so relative translational motion can occur; rolling requires additional contact physics not specified.

5 INTERPRET

Literal conclusion: relative sliding/translation is explainable; rolling is under-specified.

ANSWER

Rolling is not justified by the literal smooth-floor model; the inertia comparison is the usable idea.

QUICK CHECK

Adding a specified friction force would change the rotational conclusion.

CHECK IT ANOTHER WAY

This is a source-quality ambiguity, not a license to invent a torque.

WATCH OUT

Do not silently “repair” ambiguous source wording into a new scored law statement.

REVIEW THIS IDEA

Core 1: NLM 01-03

Q11 - Solution: Light and heavy rifle recoil

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

Two rifles fire bullets with the same forward momentum. One rifle has half the mass of the other. Compare their recoil momenta and recoil speeds.



WHAT DO WE KNOW?

same bullet momentum; rifle masses M and 2M.

REASONING LADDER

1 UNDERSTAND

Each rifle receives the same recoil momentum magnitude opposite its bullet.

2 REPRESENT

Use $p_{rifle}=-p_{bullet}$ and $v=p/M$.

3 CONNECT

$v_{light}=|p|/M$; $v_{heavy}=|p|/(2M)$.

4 CALCULATE

Therefore $v_{light}=2v_{heavy}$ while the recoil momenta are equal in magnitude.

5 INTERPRET

Lighter rifle recoils twice as fast; recoil momentum magnitudes are equal.

ANSWER

Equal recoil momentum; lighter rifle has twice the recoil speed.

QUICK CHECK

$M_{light}\,v_{light} = M_{heavy}\,v_{heavy} = |p|$.

CHECK IT ANOTHER WAY

A larger recoil speed does not mean larger recoil momentum here.

WATCH OUT

Do not confuse equal momentum with equal velocity.

REVIEW THIS IDEA

Core 1: NLM 09-10

Q12 - Solution: Horse and cart at constant speed

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A horse pulls a cart at constant speed on a level road. The horse pulls forward on the cart with 500 N. What must the total backward resistance on the cart be?



WHAT DO WE KNOW?

cart: constant speed; horse-on-cart = 500 N forward.

REASONING LADDER

1 UNDERSTAND

Constant speed in a straight line means a=0.

2 REPRESENT

Cart FBD only; do not add cart-on-horse to the cart FBD.

3 CONNECT

Newton II gives $\Sigma F_x = 0$.

4 CALCULATE

500-R=0, so R=500 N backward.

5 INTERPRET

The resistance balances the horse's forward pull on the cart.

ANSWER

500 N backward.

QUICK CHECK

Any other resistance would make the cart accelerate, contradicting constant speed.

CHECK IT ANOTHER WAY

The horse-cart third-law pair is not the same as the balanced forces on the cart.

WATCH OUT

Do not cancel forces that act on different bodies.

REVIEW THIS IDEA

Core 1: NLM 01-03, NLM 08

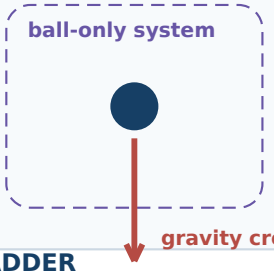
Q13 - Solution: Why a falling ball alone does not conserve momentum

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A ball falls freely and its downward momentum increases. Can you say the ball's momentum is conserved because the initial and final momentum happen to have some matching magnitude in two observations?



WHAT DO WE KNOW?

system: ball only; gravity acts throughout fall.

REASONING LADDER

1 UNDERSTAND

Gravity changes the ball momentum continuously.

2 REPRESENT

Ball-only boundary has external gravitational force.

3 CONNECT

Momentum conservation for the ball alone is not valid over the fall interval.

4 CALCULATE

Any equality of selected momentum magnitudes would be coincidental, not a conservation proof.

5 INTERPRET

Ball-only momentum is not conserved during free fall.

ANSWER

No; gravity gives external impulse to the ball-only system.

QUICK CHECK

Including Earth + ball can move the gravitational interaction inside the system boundary.

CHECK IT ANOTHER WAY

Conservation is about the system and external impulse, not just matching numbers.

WATCH OUT

Do not infer conservation from one object's before/after momentum values.

REVIEW THIS IDEA

Core 1: NLM 09

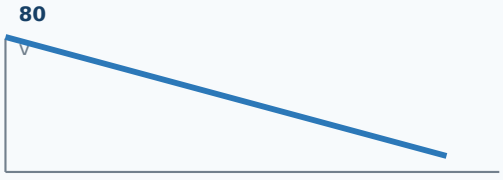
Q14 - Solution: Friction from a velocity-time graph

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A 50 g slider slows uniformly from 80 m/s to rest in 8 s. If friction is the only horizontal force, find the friction force.



slope = acceleration → friction force if friction is the only h

WHAT DO WE KNOW?

m=0.050 kg; v changes 80→0 m/s over 8 s.

REASONING LADDER

1 UNDERSTAND

Uniform slowing means constant negative acceleration.

2 REPRESENT

Read a from the v-t slope.

3 CONNECT

a=(0-80)/8=-10 m/s².

4 CALCULATE

F=ma=0.050(-10)=-0.50 N.

5 INTERPRET

Negative sign means friction opposes the positive motion.

ANSWER

0.50 N opposite the motion.

QUICK CHECK

Impulse magnitude 0.50×8=4 N·s equals |Δp|=0.050×80=4 kg·m/s.

CHECK IT ANOTHER WAY

Graph area is displacement, not acceleration.

WATCH OUT

Do not use the area under v-t when the requested bridge is slope → acceleration → force.

REVIEW THIS IDEA

Core 1: NLM 11

Q15 - Solution: Force-mass scaling

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

An object accelerates at a under net force F and mass m. The force becomes F/2 while the mass becomes 2m. What is the new acceleration?

state	Fnet	mass	acceleration
old	F	m	a
new	F/2	2m	?

WHAT DO WE KNOW?

initial: F,m,a; new: F/2,2m.

REASONING LADDER

1 UNDERSTAND

Acceleration depends on both net force and mass.

2 REPRESENT

Use symbolic old/new states.

3 CONNECT

$a_2=(F/2)/(2m)=F/(4m)$.

4 CALCULATE

Since $a_1=F/m$, $a_2=a_1/4$.

5 INTERPRET

The acceleration becomes one quarter of the original.

ANSWER

a/4.

QUICK CHECK

If only the force halved, it would be a/2; the mass change halves it once more.

CHECK IT ANOTHER WAY

Direct substitution gives $(F/2)/(2m)=F/(4m)$, confirming one-quarter of F/m.

WATCH OUT

Do not change only one factor when both force and mass change.

REVIEW THIS IDEA

Core 1: NLM 04

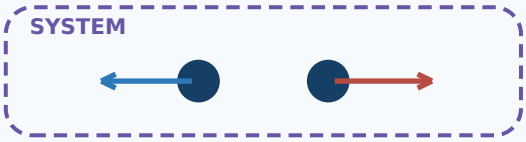
Q16 - Solution: Skaters push apart

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

Two skaters of masses 50 kg and 75 kg push apart from rest. The 50 kg skater moves left at 3 m/s. Find the velocity of the 75 kg skater.



Check external impulse before conserving total momentum.

WHAT DO WE KNOW?

system: both skaters; $p_{initial}=0$; $m_1=50$ kg, $v_1=-3$ m/s; $m_2=75$ kg.

REASONING LADDER

1 UNDERSTAND

Internal push forces exchange equal/opposite momentum; external horizontal impulse is neglected.

2 REPRESENT

Choose right positive and keep signed velocities.

3 CONNECT

$0=50(-3)+75v_2$.

4 CALCULATE

$75v_2=150$, so $v_2=+2$ m/s.

5 INTERPRET

The heavier skater moves more slowly to carry the same opposite momentum magnitude.

ANSWER

2 m/s to the right.

QUICK CHECK

$50\times3=150$ and $75\times2=150$ kg·m/s, so momenta cancel.

CHECK IT ANOTHER WAY

Equal interaction forces do not imply equal speeds.

WATCH OUT

Do not drop the sign of the left-moving skater.

REVIEW THIS IDEA

Core 1: NLM 09-10

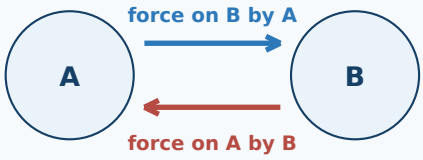
Q17 - Solution: Sprinkler reaction

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A sprinkler ejects water tangentially backward from its nozzles. Why does the sprinkler rotate in the opposite sense?



WHAT DO WE KNOW?

interaction: sprinkler ↔ expelled water.

REASONING LADDER

1 UNDERSTAND

Water is accelerated one way; the sprinkler receives the reaction force the other way.

2 REPRESENT

Use two-body interaction arrows, not a single-body cancellation.

3 CONNECT

Newton III gives equal magnitude, opposite direction forces on different bodies.

4 CALCULATE

The reaction forces on the arms create a turning effect opposite the water-jet direction.

5 INTERPRET

The sprinkler rotates opposite the ejected-water direction.

ANSWER

Opposite to the water jets.

QUICK CHECK

Reversing nozzle directions reverses the reaction and rotation sense.

CHECK IT ANOTHER WAY

The water force and sprinkler reaction do not cancel on one body.

WATCH OUT

Do not call the two arrows “balanced forces” on the sprinkler.

REVIEW THIS IDEA

Core 1: NLM 08

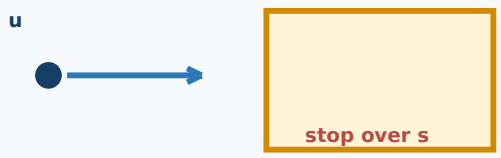
Q18 - Solution: Bullet stopping force and time

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A 0.020 kg bullet enters a block at 100 m/s and stops uniformly in 0.10 m. Find the average stopping force and stopping time.



WHAT DO WE KNOW?

m=0.020 kg; u=100 m/s; v=0; s=0.10 m; uniform deceleration.

REASONING LADDER

1 UNDERSTAND

First determine the kinematic acceleration, then convert it to force.

2 REPRESENT

Use $v^2=u^2+2as$ because time is not needed for the first step.

3 CONNECT

$0=10000+0.20a \rightarrow a=-50000$ m/s².

4 CALCULATE

$F=ma=0.020(-50000)=-1000$ N. Then $0=100+(-50000)t \rightarrow t=0.0020$ s.

5 INTERPRET

Force is 1000 N opposite the bullet motion; stopping time is 2.0 ms.

ANSWER

1000 N opposite motion; 0.0020 s.

QUICK CHECK

Impulse: $1000\times0.002=2$ N·s = $m\Delta v = 0.020\times100$.

CHECK IT ANOTHER WAY

Using impulse, $|F|t=1000(0.002)=2$ N·s, matching $m|\Delta v|=0.020(100)=2$ N·s.

WATCH OUT

Do not try to infer force directly from stopping distance without finding acceleration.

REVIEW THIS IDEA

Core 1: NLM 05-07

Q19 - Solution: Combined masses under the same force

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

A 5 N force gives mass m_1 acceleration 8 m/s² and mass m_2 acceleration 24 m/s². If the masses are tied together and the same 5 N net force acts, find the acceleration.

case	F	a	m=F/a
1	5 N	8 m/s²	5/8 kg
2	5 N	24 m/s²	5/24 kg

WHAT DO WE KNOW?

F=5 N; a_1=8 m/s²; a_2=24 m/s²; tied system under same 5 N.

REASONING LADDER

1 UNDERSTAND

Infer individual masses from the same force.

2 REPRESENT

m_1=5/8 kg; m_2=5/24 kg; total mass is their sum.

3 CONNECT

m_total=5/8+5/24=15/24+5/24=20/24=5/6 kg.

4 CALCULATE

a=F/m_total=5/(5/6)=6 m/s².

5 INTERPRET

The tied pair accelerates at 6 m/s².

ANSWER

6 m/s².

QUICK CHECK

The result is below both individual accelerations because the combined mass exceeds either mass.

CHECK IT ANOTHER WAY

1 N = 1 kg·m/s²; keep SI units consistent.

WATCH OUT

Do not average the two accelerations; add masses, then use F/m_total.

REVIEW THIS IDEA

Core 1: NLM 04

Q20 - Solution: Momentum graphs and force as rate of change

Complete reasoning ladder - no skipped learner steps

SOLUTION

QUESTION RECAP

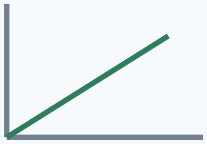
For p=mv, what do the slopes of a p-v graph at fixed mass and a p-m graph at fixed velocity represent? How is force related to momentum change?

p vs v



slope = mass

p vs m



slope = velocity

WHAT DO WE KNOW?

relation p=mv; compare p-v and p-m graphs; force from momentum rate.

REASONING LADDER

1 UNDERSTAND

Both graphs are linear through the origin when the held quantity is constant.

2 REPRESENT

p-v: slope m. p-m: slope v.

3 CONNECT

Force measures the time rate of momentum change.

4 CALCULATE

Thus slope(p-v)=m, slope(p-m)=v, and $F_{avg}=\Delta p/\Delta t$.

5 INTERPRET

The graph slopes have direct physical meaning and force is not itself momentum.

ANSWER

slope(p-v)=m; slope(p-m)=v; $F_{avg}=\Delta p/\Delta t$.

QUICK CHECK

Units: (kg·m/s)/(m/s)=kg for the p-v slope; (kg·m/s)/kg=m/s for the p-m slope.

CHECK IT ANOTHER WAY

Dimensional check: slope(p-v) has units kg; slope(p-m) has units m/s.

WATCH OUT

Do not read graph height as the slope or confuse momentum with force.

REVIEW THIS IDEA

Core 1: NLM 06-07